Tribhuvan University

Faculty of Humanities & Social Sciences

OFFICE OF THE DEAN

2024

Bachelor in Computer Applications

Course Title: **Mathematics I**

Code No: Semester: **I**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Note: *Group A (Multiple Choice Questions) is not available from the source. The paper begins with Group B.*

Group B

2. Solve the inequality $3 + 2x - x^2 \geq 0$.

3. Find the domain and range of the function $f(x) = \sqrt{6 - x - x^2}$.

4. If a, b, c, and d are in G.P., prove that $a^2 - b^2, b^2 - c^2, c^2 - d^2$ are also in G.P.

5. Prove that $$\begin{vmatrix} 1 + x & 1 & 1 \\ 1 & 1 + y & 1 \\ 1 & 1 & 1 + z \\ \end{vmatrix} = xyz \left(\frac{1}{x} + \frac{1}{y} + \frac{1}{z} \right)$$

6. Find the equation of the ellipse whose latus rectum is 5 and the eccentricity is $\frac{1}{\sqrt{2}}$.

7. If $\vec{a} = \sqrt{3} \hat{i} + \hat{j}$ and $\vec{a} \times \vec{b} = (1, 2, 2)$, find the angle between $\vec{a}$ and $\vec{b}$.

8. How many numbers of three different digits less than 500 can be formed from the integers 1, 2, 3, 4, 5, and 6?

Group C

Attempt any TWO questions[2x10=20]

9. Prove that $$\frac{3 + 4i}{1 - i} + \frac{3 - 4i}{1 + i}$$ is a real number.b. If $x^2 + y^2 = 11xy$, prove that $$\log\left(\frac{x - y}{3}\right)^2 = \frac{1}{2} (\log x + \log y)$$

10. a.Find the Maclaurin series of the function $f(x) = \cos x$.b.Take any matrix of order 3 × 3 and express it as a sum of symmetric and skew-symmetric matrix.

11. a.Find the equation of a hyperbola in standard form having focus (-2, 0) and Directrix $x = -\frac{1}{2}$.b.In an examination paper on mathematics, 20 questions are set. In how many different ways you can choose 18 questions to answer?